Software Requirement Specification

E-Commerce Order Management Database System

Project Phase: Week 1 – Understanding the Business Problem

Prepared By: Database Analyst (Student Project)

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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) defines the functional and non-functional requirements for the E-Commerce Order Management Database System. It is intended to guide the database design, normalization, SQL implementation, and reporting phases of the project and to serve as the agreed reference between the business scenario and the technical solution.

1.2 Scope

The system will provide a centralized relational database to manage customer registration, product and category catalogs, supplier records, order processing, payment tracking, shipment tracking, and customer reviews for an online shopping platform. The system is limited to the data management layer; it does not cover front-end application development or third-party payment/logistics API integration.

1.3 Intended Audience

- Database designers and developers implementing the schema.
- Business stakeholders reviewing requirement coverage.
- Instructors/evaluators assessing the semester project deliverables.

1.4 Definitions and Abbreviations

Term	Definition
SRS	Software Requirement Specification
ER	Entity-Relationship (diagram/model)
ID	Unique Identifier (primary key) for an entity
FK	Foreign Key — a field linking one entity to another
DBMS	Database Management System

2. Overall Description

2.1 Product Perspective

The system is a standalone database supporting an e-commerce platform's back-office operations. It centralizes previously fragmented information (customers, products, orders, payments, shipments, reviews) into one structured, normalized data store.

2.2 Product Functions (Summary)

- Customer registration and profile management.
- Product catalog and category management.
- Supplier record management.
- Order placement and order line-item tracking.
- Payment record tracking.
- Shipment and delivery status tracking.
- Customer product review and rating management.
- Business reporting on sales, orders, and product performance.

2.3 User Classes and Characteristics

- **Customers** – interact indirectly through the platform to register, order, and review.
- **Administrators/Staff** – manage catalog, inventory, orders, payments, and shipments.
- **Database Administrator** – maintains schema integrity, security, and performance.
- **Management** – consumes reports for decision-making.

2.4 Operating Environment

The system is designed to run on a standard relational Database Management System (DBMS) accessible to the application layer of the e-commerce platform.

3. Data Model Overview

The system is built on ten mandatory core entities, listed below with their defined attributes. These entities and attributes are fixed for the project and must not be added to, removed, or renamed.

Entity	Attributes
Customer	Customer ID, Name, Email, Mobile Number, Address, Registration Date
Category	Category ID, Category Name, Description
Product	Product ID, Product Name, Price, Stock Quantity, Category ID
Supplier	Supplier ID, Supplier Name, Contact Information
Order	Order ID, Customer ID, Order Date, Total Amount, Order Status
Order Details	Order Detail ID, Order ID, Product ID, Quantity, Unit Price
Payment	Payment ID, Order ID, Payment Method, Payment Date, Payment Status
Shipment	Shipment ID, Order ID, Delivery Address, Shipment Date, Delivery Status
Review	Review ID, Customer ID, Product ID, Rating, Comments

Key relationships: A Customer places many Orders; an Order contains many Order Details, each referencing one Product; a Product belongs to one Category; an Order has one Payment and one Shipment; a Customer may write many Reviews, each tied to one Product.

4. Functional Requirements

ID	Requirement
FR-1	The system shall register customers with unique Customer ID and Email.
FR-2	The system shall organize products under categories.
FR-3	The system shall maintain supplier records for procurement reference.
FR-4	The system shall allow an order to include multiple products with quantity and unit price.
FR-5	The system shall calculate order total amount from its order details.
FR-6	The system shall record payment method, date, and status per order.
FR-7	The system shall record shipment address, date, and delivery status per order.
FR-8	The system shall allow customers to submit ratings and comments for products.
FR-9	The system shall support generation of sales, order-status, and product-performance reports.
FR-10	The system shall update product stock quantity as orders are placed.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Performance	Queries on orders/products/customers should return promptly as data grows.
NFR-2	Scalability	Schema should support increasing customers, products, and orders.
NFR-3	Data Integrity	Referential integrity must be enforced across related entities.
NFR-4	Security	Customer and payment data must be protected from unauthorized access.
NFR-5	Reliability	Order, payment, and shipment data must stay consistent during failures.
NFR-6	Maintainability	Schema should be normalized and documented for easy maintenance.
NFR-7	Backup/Recovery	Data should be recoverable through periodic backups.

6. External Interface Requirements

- **User Interfaces:** Not covered in this database-only phase; assumed to be provided by a separate front-end application.
- **Database Interfaces:** Standard SQL interface for CRUD operations and reporting queries.
- **Hardware Interfaces:** None specific; standard server/DBMS hosting environment assumed.

7. Assumptions, Dependencies, and Constraints

- The ten core entities and their attributes are fixed and must be used exactly as defined.
- Each order contains at least one order detail (product line item).
- Each order is expected to have at most one payment and one shipment record under normal operation.
- A review can only reasonably be submitted by a customer for a product, independent of order verification at this schema stage.
- The project depends on a relational DBMS being available for implementation in later phases.

8. Appendix – Traceability to Business Scenario

This SRS traces directly back to the business scenario and core entities supplied for the E-Commerce Order Management Database System project, and is consistent with the accompanying Requirement Analysis Report and Business Requirement Document produced in this phase.